

2-Kernels of a Digraph

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1 2-Kernel

Definition 1.1 (Independent Set). Let $D = (V, A)$ be a digraph. A set $S \subseteq V(D)$ is **independent** if no two vertices of S are joined by an arc of D , i.e. $(u, v) \notin A(D)$ for every $u, v \in S$.

Definition 1.2 (2-Dominant Set). Let $D = (V, A)$ be a digraph. A set $S \subseteq V(D)$ is **2-dominant** if every vertex not in S has at least two out-neighbours in S , i.e. for every $x \in V(D) \setminus S$,

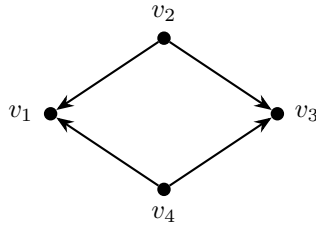
$$|\{y \in S : (x, y) \in A(D)\}| \geq 2.$$

Definition 1.3 (2-Kernel). Let $D = (V, A)$ be a digraph. A subset $N \subseteq V(D)$ is a **2-kernel** of D if N is both independent and 2-dominant. Equivalently, N is a 2-kernel if

1. no two vertices of N are joined by an arc of D , and
2. every vertex $x \in V(D) \setminus N$ has arcs into at least two distinct vertices of N .

This is stronger than the classical (von Neumann–Morgenstern) kernel, which only requires each outside vertex to reach N by *some* arc; a 2-kernel requires *at least two*.

Example 1.4. In the digraph below, $N = \{v_1, v_3\}$ is a 2-kernel: v_1 and v_3 are not joined by an arc, and each of v_2, v_4 has an arc into both v_1 and v_3 , so each outside vertex has two arcs into N .



Remark 1.5. If N is a 2-kernel of D , then N is also an ordinary kernel of D (independence is unchanged, and having two arcs into N certainly gives at least one). The converse fails: $N = \{v_1, v_3\}$ ceases to be a 2-kernel if, say, the arc $v_4 \rightarrow v_3$ is deleted, even though it remains an ordinary kernel as long as $v_4 \rightarrow v_1$ survives.